

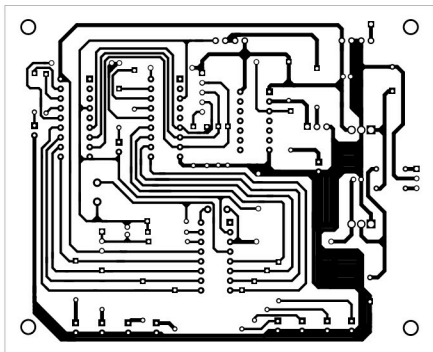


Fig. 6: An actual-size PCB pattern of the receiver unit

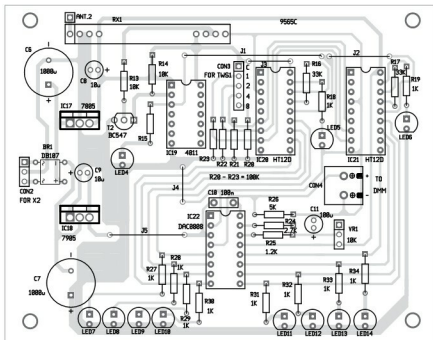


Fig. 7: Component layout of the PCB shown in Fig. 6

### STATUS OF LEDs IN THE RECEIVER UNIT

| LED      | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14  |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|
| Data bit | DB7 | DB6 | DB5 | DB4 | DB3 | DB2 | DB1 | DB0 |
| Weight   | 128 | 64  | 32  | 16  | 8   | 4   | 2   | 1   |
| Status   | Off | On  | Off | Off | On  | On  | Off | Off |

at the input and see that the multimeter shows it correctly. If required, re-adjust the trim potmeter. After proper calibration, enclose the circuit in two separate boxes with suitable connections of input and LED indicators.

### How to use

1. Attach LM35 sensors (IC9 through IC16) of the transmitter unit to different subjects of interest, noting inputs (IN0-IN7) of the ADC to which these are connected individually.

2. Attach a digital multimeter to the output of the receiver unit with positive lead to ground and negative lead to R26. Set the multimeter's dial to 200mV range.

3. Switch on both transmitter and receiver units.

4. Rotate the rotary thumbwheel switch to get the temperature of a particular subject. For example, if you like to get temperature of the subject attached to the sensor connected to IN6 input of the ADC, the thumbwheel switch ought to be rotated to position 6.

5. A temperature of 27.5°C would be displayed as 27.5mV (when value of R26 is around 500-ohm). If R26 is 5k, displayed value would be 0.275V when DMM range switch is set to 2V.

### Applications

Although the system can be used best to measure temperatures in hazardous or inaccessible areas (like a radioactive zone), the same can also be used by a hospital doctor to monitor, from a fixed location, the body temperatures of multiple patients lying in different rooms without visiting each patient in person.

A hotel control room can monitor temperatures of all the rooms at the same time by using multiple units. The unit can also be used (with certain modifications) as a wireless digital voltmeter. **DAV**

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